SAMPLE

Earth's Sky: Sundials and Shadow Clocks

Ancient Sky-Based Timekeeping

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Free sample: Introduction & Chapter 1

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Introduction

Time, for nearly all of human history, was something the sky kept. The oldest instruments that read it were not machines at all but arrangements of stick, stone, and shadow, and the present volume is their story: the sundials and shadow clocks that carried civilization from the first marked noon to the hour before the pendulum. The subject has a claim on any reader's attention that goes beyond antiquarian charm. The shadow clock was the first astronomical instrument, the first device that turned the sky's rotation into a number a person could use, and every later clock, from the water wheel to the quartz crystal, has been an attempt to carry that shadow's authority indoors. The story of how the shadow was tamed, graduated, corrected, and finally retired is the story of humanity's long negotiation with the day itself.

The book follows the pattern of this series, sixteen rooms in the canonical house. The first chapters trace the discovery of the shadow clock, from the Egyptian shadow boards of the second millennium BC to the refined dials of the scientific revolution. The middle chapters explain what a sundial actually is, how its geometry earns its readings, and why a well-made dial in a garden can be trusted while a mass-produced one cannot. The later chapters set the instrument in its context, follow the scholars who turned dial-making into a mathematical science, and close with the puzzles that remain, the directions the craft still travels, and the numbers and stories that make the subject such a delight to carry. The aim throughout is a plain account a curious reader can follow without trigonometry, while the underlying mathematics is named honestly where it appears.

A warning about honesty belongs up front, because the sundial is surrounded by a small industry of misinformation. The instrument does not fail because it is primitive; it fails, when it fails, because it was built for a different latitude than the one it stands in, or because nobody applied the corrections every honest dial deserves. The equation of time, the longitude correction, the seasonal tilting of the sun's path: these are not defects of the instrument but the actual shape of solar time, which clocks average away. To read a sundial properly is to understand the difference between the sun's honest account and the state's smoothed one, and that understanding is itself a small education in astronomy. The reader who finishes this volume should be able to face any garden dial and know, within a minute or two, what time it is truly telling.

Chapter 1: The Discovery of Sundials and Shadow Clocks

The discovery was no single event but a slow recognition, repeated across civilizations, that a shadow's length and direction could be made to speak. The earliest surviving timepieces of this kind are the shadow clocks of ancient

Egypt and Babylon, built around fifteen centuries before the common era. These were not the graduated dials of later ages but simple boards, marked with a row of hour sectors, set up so that the shadow of a crossbar crept along the scale as the day wore on. The Egyptian versions, of which museum examples survive, worked as a leveled L: a horizontal base with a raised crossbar at one end, oriented east and west so the morning shadow fell westward and shortened toward the crossbar at midday. The device divided the day into chunks that shifted with the season, because the sun rises and sets at different points through the year, but it gave the priesthood of the Nile a workable calendar of the daylight, and that was the point.

The Mesopotamian cousins of the Egyptian board worked by shadow length rather than shadow direction. Babylonian and Assyrian texts tabulate the ratio of a vertical rod's shadow to its height at successive points of the morning and afternoon, a scheme that survives in the curious fact that some ancient cultures reckoned the day's passing by the shrinking and growing of a man's own silhouette. Length-based shadow reckoning needs only a stick and a memory for numbers, and it traveled wherever a vertical object stood: the word gnomon, which modern dial makers use for any shadow-throwing edge, is Greek for the marker of a sundial, and the Greeks themselves credited the instrument's arrival, variously, to sages of the east.

Egypt's obelisks enlarged the principle to monumental scale. A tall tapering monolith in a temple court threw a sweeping morning shadow to the west and an afternoon shadow to the east, and the court's pavement became a dial face on which priests could read the festivals. Whether the obelisks carried formal hour marks is debated, but their role as calendar keepers is not: the shadow's extreme northern and southern swings at the solstices, and the days when the shadow tip crossed particular lines, marked the turning points of the year, and Egyptian builders oriented whole temple complexes with an eye to such shadow events. The obelisk is thus the ancestor of every public clock tower, a timepiece too large to ignore and too slow to be wrong by much.

The Greek world took the shadow clock and made it mathematical. By the third century before the common era,

Eratosthenes had used the shadow of an obelisk at Syene, together with the geometry of a deep well whose bottom the sun lit only at midsummer noon, to measure the circumference of the earth; the story belongs properly to the history of geodesy, but it shows how far the humble shadow had come, from a crude hour counter to the probe of a planet's size. A few centuries later Ptolemy tabulated the latitudes of cities from the angle the noon shadow made, and the shadow had become an instrument of geography as well as of

time. The Greeks also invented the hemispherical dial, a bowl with a shadow-casting vane whose inner surface carried a web of hour curves carved for the seasons, and the conical dial that simplified the bowl's curves; both types appear in the Roman marketplaces that copied Greek practice.

Rome inherited the whole apparatus and catalogued it. Vitruvius, the architect writing in the decades around the common era's start, lists in his treatise on architecture the dial types known in his day, naming them for their inventors: the discus in plane, the hemicyclium excavated from a block, the spider's-web dial of Eudoxus, the arachne, the conical dial of Dionysodorus, and others, a dozen named families. His book also preserves a social history in a sentence: when a dial brought from Catania was set up in Rome, it told the hours of Catania, and the mistake went unnoticed for years, which says something about how loosely a city that owned the Mediterranean still held the idea of latitude. The Tower of the Winds in Athens, built a hundred years before Vitruvius, carried on its octagonal faces a whole portfolio of dials, and inside it a water clock served the cloudy days; the pairing is the oldest known public installation of shadow time with a mechanical backup.

After the Roman collapse, the shadow clock found its second career in the church. The canonical hours, the fixed pattern of daily prayer, needed marking through the daylight, and from the seventh to the fourteenth century monasteries and parish churches carried scratch dials, small vertical or horizontal faces cut into the stone of south walls, telling the hours of the office rather than equal hours of the clock. These mass dials survive by the thousand on English and continental churches, a quiet census of medieval timekeeping. When the mechanical clock began its rise in the fourteenth century the two instruments shared the field for centuries, and the printed treatise kept the older craft honest: Giovanni Padovani published instructions for making and setting out wall and horizontal dials in 1570, Giuseppe Biancani wrote on the construction of a perfect dial around 1620, and by the later sixteenth century the sundial had become the common instrument of every town square that could afford one, a status it held until the middle of the seventeenth century, when the pendulum began to dispute its authority.

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